

## ALPNACH AP, Switzerland

WMO: 066580

Lat: 46.943N

Lon: 8.286E

Elev: 1460

StdP: 13.94

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 18.0       | 22.7 | 12.4                          | 10.9 | 19.4 | 17.5 | 14.1 | 23.9 | 18.1                  | 32.6 | 16.1 | 31.2 | 6.7                   | 20   | 0.489 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.9                   | 85.8            | 70.1 | 82.3 | 68.9 | 78.9 | 67.0 | 71.0                | 82.7 | 69.5 | 80.2 | 67.8 | 77.0 | 8.4                  | 60   |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.6                            | 103.3 | 77.2 | 64.8 | 97.1 | 75.0 | 64.2 | 95.0 | 74.1 | 35.8          | 83.2 | 34.3 | 80.6 | 33.1 | 77.5 | 79.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |
| Min               | Max  |      | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |
| 15.6              | 13.6 | 11.9 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | DB                         |     |                    |     |                                                    |     |            |     |            |     |            |     |
|                   |      |      | WB                         |     |                    |     |                                                    |     |            |     |            |     |            |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 50.4   | 35.2  | 34.4  | 42.2  | 50.2  | 56.6  | 64.6  | 66.8  | 65.8  | 59.5  | 50.9  | 41.9  | 36.1  |
|                                                                        | DBStd       | 13.16  | 6.25  | 6.84  | 6.05  | 6.37  | 6.28  | 5.61  | 5.32  | 5.69  | 5.13  | 5.88  | 5.79  | 6.07  |
|                                                                        | HDD50       | 1976   | 459   | 436   | 247   | 74    | 14    | 0     | 0     | 0     | 1     | 63    | 250   | 432   |
|                                                                        | HDD65       | 5585   | 924   | 856   | 708   | 444   | 269   | 74    | 43    | 61    | 178   | 437   | 694   | 897   |
|                                                                        | CDD50       | 2132   | 1     | 0     | 5     | 80    | 217   | 438   | 520   | 489   | 285   | 91    | 6     | 0     |
|                                                                        | CDD65       | 266    | 0     | 0     | 0     | 0     | 8     | 62    | 99    | 86    | 11    | 0     | 0     | 0     |
|                                                                        | CDH74       | 2102   | 0     | 0     | 0     | 18    | 93    | 546   | 750   | 632   | 63    | 0     | 0     | 0     |
|                                                                        | CDH80       | 525    | 0     | 0     | 0     | 0     | 10    | 152   | 198   | 163   | 2     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 4.0    | 3.7   | 3.9   | 4.3   | 4.9   | 4.8   | 4.7   | 4.4   | 3.8   | 3.6   | 3.3   | 3.3   | 3.3   |
| (15) Precipitation                                                     | PrecAvg     | 64.8   | 3.7   | 4.1   | 4.3   | 5.0   | 6.9   | 7.2   | 7.2   | 7.4   | 5.4   | 4.2   | 4.9   | 4.6   |
|                                                                        | PrecMax     | 81.5   | 8.8   | 12.1  | 8.5   | 10.0  | 10.9  | 11.1  | 13.7  | 16.0  | 9.2   | 7.5   | 13.6  | 11.4  |
|                                                                        | PrecMin     | 49.3   | 0.2   | 1.3   | 1.7   | 1.1   | 1.7   | 4.3   | 3.1   | 2.2   | 2.1   | 0.2   | 0.2   | 0.2   |
|                                                                        | PrecStd     | 7.3    | 2.2   | 2.4   | 1.9   | 2.0   | 2.8   | 1.7   | 2.6   | 2.9   | 1.8   | 1.8   | 2.7   | 2.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 55.5  | 56.8  | 64.8  | 76.7  | 81.7  | 89.5  | 89.4  | 88.1  | 79.0  | 69.8  | 61.9  | 54.0  |
|                                                                        |             | MCWB   | 48.5  | 46.4  | 51.8  | 57.8  | 65.1  | 73.6  | 71.5  | 71.0  | 67.5  | 60.2  | 53.5  | 47.3  |
|                                                                        | 2%          | DB     | 50.2  | 52.0  | 61.2  | 71.8  | 76.7  | 84.2  | 85.7  | 84.2  | 75.4  | 66.2  | 57.1  | 50.4  |
|                                                                        |             | MCWB   | 45.2  | 44.3  | 50.2  | 56.6  | 63.1  | 69.5  | 69.7  | 70.0  | 65.6  | 58.4  | 51.0  | 45.6  |
|                                                                        | 5%          | DB     | 46.6  | 48.1  | 57.6  | 68.2  | 73.1  | 80.2  | 82.1  | 80.8  | 71.8  | 62.8  | 53.5  | 46.8  |
|                                                                        |             | MCWB   | 43.1  | 41.8  | 48.3  | 55.5  | 60.3  | 67.6  | 68.7  | 68.8  | 63.0  | 56.5  | 49.4  | 43.8  |
|                                                                        | 10%         | DB     | 44.3  | 44.4  | 53.9  | 64.0  | 69.4  | 75.6  | 78.6  | 77.1  | 69.4  | 60.5  | 50.3  | 44.6  |
|                                                                        |             | MCWB   | 41.7  | 40.4  | 46.7  | 53.1  | 58.8  | 65.1  | 66.7  | 67.0  | 62.2  | 55.9  | 46.8  | 42.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 49.5  | 48.2  | 53.4  | 60.0  | 66.5  | 73.9  | 72.9  | 72.6  | 68.2  | 61.7  | 54.4  | 49.2  |
|                                                                        |             | MCDB   | 54.3  | 55.1  | 62.5  | 72.5  | 78.0  | 87.1  | 85.8  | 84.1  | 77.2  | 67.8  | 60.2  | 53.1  |
|                                                                        | 2%          | WB     | 45.9  | 44.6  | 50.8  | 57.8  | 63.9  | 70.5  | 70.6  | 70.8  | 65.9  | 59.2  | 51.7  | 46.6  |
|                                                                        |             | MCDB   | 49.4  | 50.2  | 59.9  | 69.9  | 74.6  | 82.1  | 82.7  | 81.9  | 73.6  | 64.2  | 55.8  | 50.0  |
|                                                                        | 5%          | WB     | 43.5  | 42.2  | 48.9  | 55.6  | 61.6  | 68.1  | 68.9  | 69.3  | 63.9  | 57.6  | 49.7  | 44.3  |
|                                                                        |             | MCDB   | 45.9  | 46.6  | 56.8  | 66.7  | 70.0  | 77.5  | 79.4  | 79.6  | 70.7  | 62.0  | 53.4  | 46.3  |
|                                                                        | 10%         | WB     | 41.2  | 40.1  | 47.1  | 53.7  | 59.3  | 66.1  | 67.4  | 67.6  | 62.3  | 55.8  | 47.6  | 42.0  |
|                                                                        |             | MCDB   | 43.0  | 43.5  | 53.8  | 63.0  | 67.3  | 75.1  | 76.6  | 76.0  | 68.4  | 59.8  | 49.9  | 43.9  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.2  | 13.1  | 18.1  | 20.1  | 17.5  | 18.1  | 17.9  | 17.2  | 16.0  | 14.2  | 12.2  | 10.5  |
|                                                                        |             | MCDBR  | 16.0  | 21.2  | 25.7  | 28.7  | 24.9  | 24.9  | 24.7  | 23.6  | 20.6  | 19.3  | 16.9  | 14.9  |
|                                                                        |             | MCWBR  | 11.6  | 13.9  | 15.9  | 16.0  | 12.7  | 12.9  | 11.6  | 12.1  | 12.2  | 12.5  | 11.5  | 10.6  |
|                                                                        | 5% WB       | MCDBR  | 15.4  | 19.5  | 24.2  | 26.9  | 22.2  | 22.7  | 22.4  | 22.3  | 19.4  | 17.6  | 16.7  | 14.6  |
|                                                                        |             | MCWBR  | 11.5  | 13.3  | 15.5  | 15.4  | 11.7  | 12.1  | 11.3  | 11.8  | 11.6  | 11.9  | 11.6  | 10.6  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.295 | 0.328 | 0.380 | 0.427 | 0.447 | 0.452 | 0.453 | 0.439 | 0.405 | 0.372 | 0.328 | 0.292 |
|                                                                        | taud        |        | 2.329 | 2.236 | 2.158 | 2.103 | 2.118 | 2.166 | 2.165 | 2.215 | 2.286 | 2.342 | 2.384 | 2.353 |
|                                                                        | Ebn at Noon |        | 250   | 262   | 264   | 262   | 261   | 260   | 257   | 256   | 253   | 244   | 235   | 237   |
|                                                                        | Edh at Noon |        | 23    | 32    | 40    | 47    | 48    | 46    | 45    | 42    | 36    | 29    | 23    | 21    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 434   | 719   | 1101  | 1394  | 1555  | 1816  | 1782  | 1550  | 1190  | 769   | 449   | 358   |
|                                                                        | RadStd      |        | 52    | 93    | 156   | 212   | 153   | 166   | 158   | 147   | 117   | 85    | 61    | 52    |

## Historical Trends

|                          | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|--------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                          |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only             | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (120 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                          | N/A   | N/A     | N/A    | +1.30   | +0.72 | +0.56 | N/A         | N/A   | +103  | +29   |

Nomenclature: See separate page